David Cellini

Senior Full-Stack Software Engineer · React, TypeScript, Node.js, GraphQL

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Open to: senior frontend or full-stack engineering roles · remote or hybrid in the North East

Profile

Senior full-stack software engineer with 10+ years building Hive, a multi-tenant employee-experience SaaS platform used by around 200 organisations. Works across React and TypeScript, GraphQL and Node.js services, and MongoDB. Has led large features end to end and small teams, and delivered platform-wide accessibility work (WCAG 2.2) and zero-downtime data migrations. Known for small, reviewable changes, careful data work and helping other engineers move faster.

Core skills

React and TypeScript · Node.js · GraphQL and Apollo · MongoDB · REST APIs · Full-stack product engineering · Design systems · Accessibility (WCAG 2.2) · Technical leadership and mentoring · AI-assisted development (Claude Code)

Experience

Senior Software Engineer · Hive HR — Jul 2024 – Present

Employee-experience SaaS for surveys, analytics and recognition (~200 organisations) · Newcastle upon Tyne, remote

- Lead engineer on the Question Bank (~120 PRs): a per-organisation MongoDB model with Mongoose discriminators, a GraphQL API and a React editor with machine translation. Migrated legacy demographic questions with zero-downtime migrations, removed ~10k lines of legacy code, and unlocked Custom Indexes and 360 Surveys.
- Built Onboarding and Exit Surveys, enforcing per-survey confidentiality across MongoDB, Elasticsearch and CSV exports, and refactored the survey builder so new metrics plug in without bespoke code.
- Rebuilt the employee homepage around a recognitions feed on MongoDB Atlas Search; one customer saw 43% of survey responses arrive through it, helping them reach a 68% response rate.
- Planned and reported WCAG 2.2 AA accessibility work: audited 97 pages, traced ~450 failures to ~80 shared-component fixes in 23 themed PRs, and shipped fixes for keyboard dismissal, focus, reflow and motion.
- Designed and built survey auto-enrolment end to end: an RFC, a new Node.js microservice, a scheduled AWS Lambda and a rule-based audience builder in React, delivered as a reviewable stack of PRs.
- Fixed production reliability issues: cut a large export's memory use from 5.8 GB to 455 MB, and turned an index-readiness race fix into a shared helper rolled out to 4 services, and fixed a live launch-day language bug the same evening. Shared the on-call rota (Sentry, CloudWatch).
- Picked up AI-generated manager reports mid-project: fixed the Python/R key-driver integration and made AWS Bedrock narrative generation run concurrently, reducing timeout-related support tickets.
- Mentored other engineers, and wrote Claude Code skills and documentation for the team on diagnosing failed deploys and local set-up.

Software Engineer · Hive HR — Sep 2016 – Jul 2024

Newcastle upon Tyne

- Led a team of 3 rebuilding survey heatmaps: a calculation library with Zod-validated contracts, a GraphQL API and a React UI. Proved parity with a comparison script before cut-over, and restored heatmaps for one of the largest customers after v1 had been switched off for them.
- Sole engineer on Scheduled Surveys: future-dated sends on AWS EventBridge, edit locking, admin reminder emails and a backfill migration. Removed the support team's manual send queue.
- Led right-to-left (Arabic) support: moved the design system to CSS logical properties with a direction provider and hook, now used by 13 customer organisations.
- Built Self-Select Demographics end to end: Elasticsearch bucketing and filtering, a dedicated access role and the React UI.
- Built much of the drag-and-drop survey builder in React and Redux, the survey and campaign dashboards, and the Engagement Index metric with a migration for historic surveys.
- 2016–2019: ran customer support single-handedly alongside engineering (survey set-ups, launches, translations and the V1 platform migration), then trained the first support hires.

Feedback Analyser · Personal project 2026

- Built an AI app in TypeScript that analyses employee feedback (sentiment, themes, priority and summaries) and answers questions using embedding-based semantic search, RAG and a multi-step tool-calling agent on the Gemini API.
- Measured output quality with an evaluation suite of deterministic checks, semantic checks and LLM-as-judge, and added grounding checks that reject any answer whose quotes, sources or numbers don't match the evidence. Redacts personal data before answers are shown.
- Validated every model response with Zod and kept the app provider-agnostic, with mock providers so the Vitest tests never call a live model. Express API and React front end, deployed on Cloudflare Workers.

Portfolio site · Personal project 2026

- Built and launched davidcellini.co.uk: a static Astro site on Cloudflare with a case study viewer that works without leaving the page, built to WCAG 2.2 AA.

Earlier experience

Web Developer · Leighton Jun 2014 – Sep 2016

Digital agency · Houghton-le-Spring

- Front-end lead on the new ATG Tickets website: organised and delegated work across the team, built the most complex components and mentored a junior developer, working in an Agile team.
- Built components and templates for the new Home Group website, as sole front-end developer for the second half of the project, plus its style guide and an 80th-anniversary microsite.
- Added new features to and maintained the British Airways Careers website, and delivered fixes and improvements across ba.com.

Digital Content Editor · GMG Radio Feb 2008 – Apr 2014

Real and Smooth Radio · Gateshead

Ran seven radio station websites: search-friendly content, video, podcasts, social media and performance reporting for internal teams.

Music Information Assistant · ExploreMusic Nov 2004 – Feb 2008

Gateshead Council · The Sage Gateshead

Kept the music library's website and its tutor and workshop database up to date, built online archives of school events and workshops for families to revisit, and taught library users everyday IT skills.

Technical skills

- **Frontend:** React, TypeScript, JavaScript, HTML, CSS/Sass, Emotion, Redux Toolkit, Apollo Client, Storybook, design systems, WCAG 2.2
- **Backend and data:** Node.js, GraphQL (Apollo), REST/OpenAPI, Express, RxJS, Zod, MongoDB, Mongoose, Atlas Search, Elasticsearch, Handlebars
- **Cloud, delivery and testing:** AWS Lambda, AWS Batch, Amazon S3, Amazon SNS, Amazon SQS, Route 53, CloudWatch, Amazon EventBridge, Sentry, Docker, GitHub Actions, Buildkite, pnpm, Git, Vitest, Jest
- **AI tools:** Claude Code, Anthropic API, Amazon Bedrock, Gemini API, GitHub Copilot, Superset
- **Practices:** internationalisation and right-to-left, feature flags, RFCs and code review, mentoring

Education and certifications

- **BA International Business (2:1)** · University of Sunderland · 2003
- **Certified Internet Webmaster: Site Development and IT Fundamentals Diploma (Distinction)** · Computeach · 2007
- **Advanced GNVQ Business Studies (Distinction)** · St Robert of Newminster Sixth Form · 2000
- **12 GCSEs including English Language (A) and Maths (B)** · St Robert of Newminster School · 1998

Awards

- **106 peer recognitions** · Hive, 2017–2026 · Thank-you awards from colleagues, mostly for delivering big features, helping and teaching teammates, and going the extra mile for customers
- **Unsung Hero Award** · GMG Radio, nominated by colleagues